Demand estimation and market analysis - air fryers

notebook 3 - costs, markups, and profit role: pricing economist, using a demand model to reason about price, cost, and profitability

given the demand slope:

$$\hat{s}'_{bt}(p_{bt}) = \hat{\beta}_{\text{price}} \cdot s_{bt}(1 - s_{bt})$$

and the first order pricing condition

$$\hat{c}_{bt} = p_{bt} + \frac{s_{bt}}{\hat{s}'_{bt}(p_{bt})}$$

```
import pandas as pd
import numpy as np
import matplotlib
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
import warnings
warnings.filterwarnings('ignore')

sns.set_theme(style='whitegrid', palette='tab10')
COLORS = sns.color_palette('tab10', 10)
BRANDS = ['chefman','cosori','cuisinart','dash','gowise usa',
          'instant_pot','ninja','nuwave','oster','ultrean']
BRAND_PALETTE = {b: COLORS[i] for i, b in enumerate(BRANDS)}

df = pd.read_csv('air_fryers_clean_brand_year.csv')
char_cols = ['compact_share','dual_basket_share','oven_style_share','rotisserie

# Re-run demand model
brand_dummies = pd.get_dummies(df['brand'], drop_first=True, prefix='brand').as
year_dummies  = pd.get_dummies(df['year'],  drop_first=True, prefix='year').ast
X = pd.concat([df[['avg_price','avg_rating']+char_cols].astype(float),
               brand_dummies, year_dummies], axis=1)
X = sm.add_constant(X)
y = df['log_brand_share'].astype(float)
model = sm.OLS(y, X).fit()
print(f"Price coefficient: {model.params['avg_price']:.4f}  (negative = OK)")

# Compute cost and markup
df['demand_slope']  = model.params['avg_price'] * df['brand_share'] * (1 - df['
df['unit_cost']     = df['avg_price'] + df['brand_share'] / df['demand_slope']
df['markup']        = df['avg_price'] - df['unit_cost']
df['average_profit']= df['brand_share'] * df['markup']
```

```
print(df[['brand','year','avg_price','unit_cost','markup','average_profit']].rc
```

```
Price coefficient: -0.0377  (negative = OK)
          brand  year  avg_price  unit_cost  markup  average_profit
0       chefman  2019      72.96      44.23   28.73            2.18
1        cosori  2019     159.99     133.42   26.57            0.02
2     cuisinart  2019     229.47     199.73   29.74            3.19
3          dash  2019      55.18      22.00   33.17            6.63
4    gowise usa  2019      83.58      46.07   37.51           10.96
5   instant_pot  2019      78.02      51.05   26.97            0.43
6         ninja  2019     112.16      79.28   32.88            6.33
7        nuwave  2019     151.10     123.67   27.43            0.88
8         oster  2019     191.94     164.47   27.47            0.92
9       ultrean  2019      80.62      52.68   27.95            1.40
10      chefman  2020      97.43      69.24   28.19            1.64
11       cosori  2020     115.65      88.87   26.78            0.23
12    cuisinart  2020     227.91     198.35   29.57            3.02
13         dash  2020      58.39      28.84   29.55            3.00
14   gowise usa  2020      86.95      54.95   32.00            5.46
15  instant_pot  2020     120.05      89.06   30.98            4.44
16        ninja  2020     140.45     108.62   31.82            5.27
17       nuwave  2020     119.90      91.07   28.84            2.29
18        oster  2020     185.60     158.66   26.94            0.39
19      ultrean  2020      76.84      45.36   31.48            4.93
20      chefman  2021      92.08      63.09   28.99            2.44
21       cosori  2021      97.42      69.91   27.51            0.97
22    cuisinart  2021     224.30     195.23   29.08            2.53
23         dash  2021      57.73      28.79   28.95            2.40
24   gowise usa  2021      84.63      54.60   30.02            3.48
25  instant_pot  2021     113.64      80.45   33.19            6.64
26        ninja  2021     166.68     134.21   32.47            5.92
27       nuwave  2021     136.04     108.06   27.97            1.42
28        oster  2021     184.62     157.41   27.20            0.65
29      ultrean  2021      76.80      46.01   30.79            4.24
30      chefman  2022      98.44      68.38   30.06            3.51
31       cosori  2022      97.87      69.23   28.64            2.09
32    cuisinart  2022     223.53     194.76   28.76            2.22
33         dash  2022      57.15      29.19   27.97            1.42
34   gowise usa  2022      86.32      58.13   28.19            1.64
35  instant_pot  2022     106.08      68.87   37.21           10.66
36        ninja  2022     156.38     123.23   33.15            6.60
37       nuwave  2022     144.99     117.49   27.50            0.95
38        oster  2022     191.49     164.22   27.26            0.71
39      ultrean  2022      78.27      49.48   28.79            2.24
40      chefman  2023      93.78      62.90   30.88            4.33
41       cosori  2023     100.41      69.99   30.42            3.88
42    cuisinart  2023     214.53     186.42   28.11            1.56
43         dash  2023      58.95      30.87   28.08            1.53
44   gowise usa  2023      95.80      67.76   28.04            1.50
45  instant_pot  2023     104.52      70.09   34.44            7.89
46        ninja  2023     151.05     117.10   33.95            7.40
47       nuwave  2023     133.10     104.98   28.12            1.57
48        oster  2023     191.52     164.25   27.26            0.71
49      ultrean  2023      78.04      50.39   27.64            1.10
```

3.1: average unit costs and market ups by brand

```
tbl = df.groupby('brand')[['avg_price','unit_cost','markup','average_profit']].
print(tbl)
```

```
              avg_price  unit_cost  markup  average_profit
brand
chefman           90.94      61.57   29.37            2.82
cosori           114.27      86.28   27.98            1.44
cuisinart        223.95     194.90   29.05            2.50
dash              57.48      27.94   29.54            2.99
gowise usa        87.45      56.30   31.15            4.61
instant_pot      104.46      71.90   32.56            6.01
ninja            145.34     112.49   32.85            6.31
nuwave           137.02     109.05   27.97            1.42
oster            189.03     161.80   27.23            0.68
ultrean           78.11      48.79   29.33            2.78
```

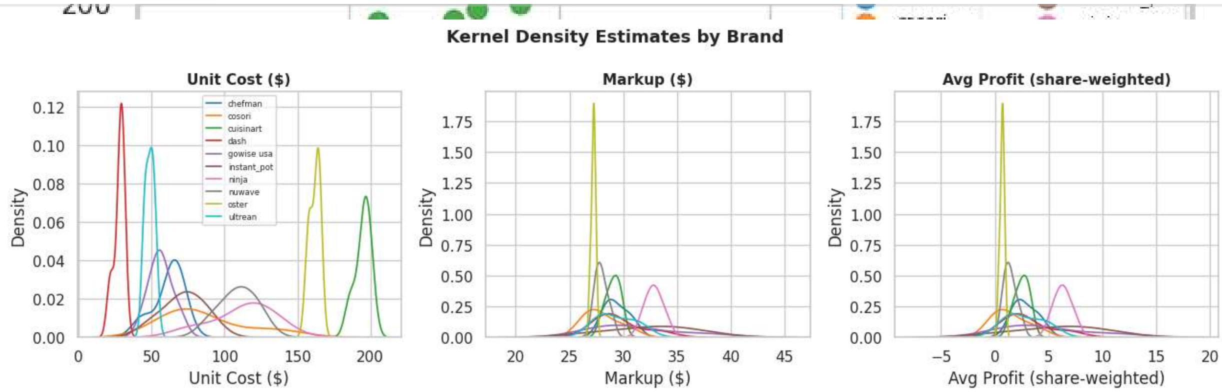
1. average unit costs and markups: Cuisinart and oster carry the highest inferred unit costs (194.90 and 161.80 respectively) consistent with their premium positioning. Dash has the lowest unit cost at 27.94 consistent with budget tier sourcing. Absolute dollar markups are relatively compressed across brands ranging from about $27 to 33$ which is economically interesting given the wide price range.
2. Are any unit costs negative? No, all inferred unit costs are positive across all brand years. This is a reassuring sanity check that the model is behaving correctly.
3. Higher cost brands and ratings? Cuisinart and oster have the highest unit costs and they also tend to have decent but not exceptional ratings (around 4.4-4.5), suggesting that higher cost does not directly translate into standout consumer satisfaction. Ninja has high unit costs (112$) and is the market leader, suggesting its cost premium is offset by strong brand equity and product quality that drives share. Dash has low costs, low prices, and competitive ratings around 4.4, showing that budget products can still satisfy consumers well.

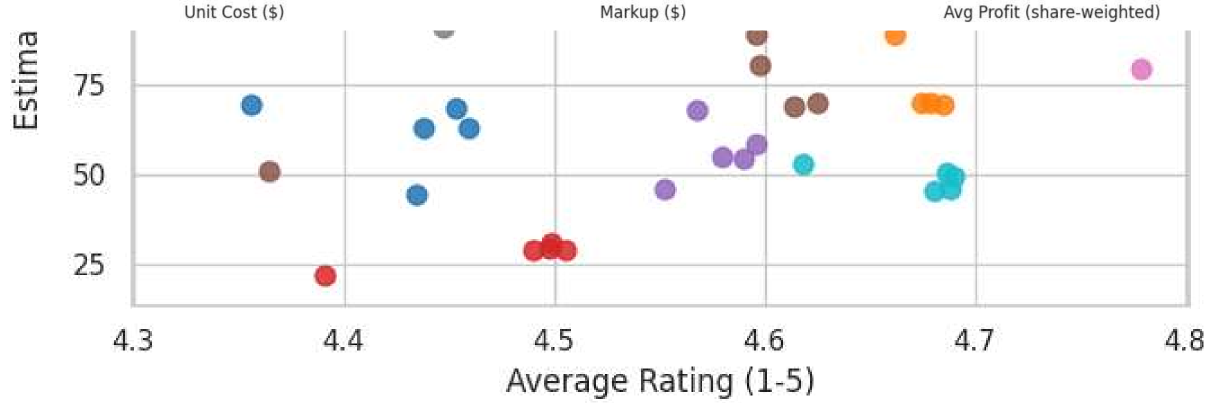
```
# Price vs Unit Cost
fig, ax = plt.subplots(figsize=(7,5))
for brand, grp in df.groupby('brand'):
    ax.scatter(grp['avg_price'], grp['unit_cost'], label=brand, color=BRAND_PAL
lo, hi = df['avg_price'].min()-5, df['avg_price'].max()+5
ax.plot([lo,hi],[lo,hi], 'k--', alpha=0.4, label='Price = Cost')
ax.set_xlabel('Average Price ($)'); ax.set_ylabel('Estimated Unit Cost ($)')
ax.set_title('Price vs Estimated Unit Cost', fontsize=13, fontweight='bold')
ax.legend(ncol=2, fontsize=7)
plt.tight_layout(); plt.show()
```

```
# Rating vs Unit Cost
fig, ax = plt.subplots(figsize=(7,5))
for brand, grp in df.groupby('brand'):
    ax.scatter(grp['avg_rating'], grp['unit_cost'], label=brand, color=BRAND_PA
ax.set_xlabel('Average Rating (1-5)'); ax.set_ylabel('Estimated Unit Cost ($)')
ax.set_title('Rating vs Estimated Unit Cost', fontsize=13, fontweight='bold')
ax.legend(ncol=2, fontsize=7)
plt.tight_layout(); plt.show()
```

Do more expensive products have higher consumer satisfaction? the rating vs unit cost scatter shows not strong positive relationship between cost and satisfication. High cost brands like cuisinart and oster do not achieve notably higher ratings than cheaper brands like chefman or ultrean. This suggests that price in this market is driven more by product segment positioning and brand prestige than by quality differences that consumers perceive and rate. Ninja is an interesting case as it commands above average unit costs and above average ratings, making itthe brand that most strongly earns its premium in consumer perception.

```
fig, axes = plt.subplots(1, 3, figsize=(13,4))
for brand, grp in df.groupby('brand'):
    c = BRAND_PALETTE[brand]
    sns.kdeplot(grp['unit_cost'],     ax=axes[0], label=brand, color=c, linewic
    sns.kdeplot(grp['markup'],        ax=axes[1], label=brand, color=c, linewic
    sns.kdeplot(grp['average_profit'],ax=axes[2], label=brand, color=c, linewic
for ax, t in zip(axes, ['Unit Cost ($)','Markup ($)','Avg Profit (share-weighte
    ax.set_xlabel(t); ax.set_title(t, fontsize=11, fontweight='bold')
axes[0].legend(fontsize=6)
fig.suptitle('Kernel Density Estimates by Brand', fontsize=13, fontweight='bolc
plt.tight_layout(); plt.show()
```





4. KDE interpretation: the unit cost distribution show clear seperation between budget brands (dash, ultrean clustered below 60), mid - tier brands ($60-100$), and

premium brands (Cuisinart, oster above $150).Themarkupdistributionsarestrikinglysimilaracrossbrands,allclus$ 27-$33 range, suggesting roughly uniform dollar markups across the market regardless of price tier. The average profit KDEs show much more seperation: ninja and instant pot have higher density mass at elevated profit values because their large market shares multiply out their moderate markups into much larger share weighted profit figures, while small share brands like oster and nuwave show compressed profit densities near zero.

3.4: average profit bar chart

```
profit_by_brand = df.groupby('brand')['average_profit'].mean().sort_values(ascen
fig, ax = plt.subplots(figsize=(8,4))
profit_by_brand.plot(kind='bar', ax=ax,
    color=[BRAND_PALETTE[b] for b in profit_by_brand.index])
ax.set_title('Average Share-Weighted Profit by Brand', fontsize=13, fontweight='
ax.set_ylabel('Avg Profit (s x m)'); ax.set_xlabel('Brand')
ax.set_xticklabels(ax.get_xticklabels(), rotation=30, ha='right')
plt.tight_layout(); plt.show()
```